

Problem 2.1: Non-overlapping wave functions



Consider a particle and two normalized energy eigenfunctions $\psi_1(\mathbf{x})$ and $\psi_2(\mathbf{x})$ corresponding to the eigenvalues $E_1 \neq E_2$. Assume that the eigenfunctions vanish outside the two non-overlapping regions Ω_1 and Ω_2 respectively.

- (a) Show that, if the particle is initially in region Ω_1 then it will stay there forever.
- (b) If, initially, the particle is in the state with wave function

$$\psi(\mathbf{x}, 0) = \frac{1}{\sqrt{2}}[\psi_1(\mathbf{x}) + \psi_2(\mathbf{x})],$$

show that the probability density $\rho = |\psi(\mathbf{x}, t)|^2$ is independent of time.

- (c) Now assume that the two regions Ω_1 and Ω_2 overlap partially. Starting with the initial wave function of case (b), show that the probability density ρ is a periodic function of time.
- (d) Starting with the same initial wave function and assuming that the two eigenfunctions are real and isotropic, take the two partially overlapping regions Ω_1 and Ω_2 to be two concentric spheres of radii $R_1 > R_2$. Compute the probability current j that flows through Ω_1 .

Hint: The probability current density is given by

$$\mathbf{j}(\mathbf{x}, t) = \frac{\hbar}{2mi} (\psi^*(\mathbf{x}, t) \nabla \psi(\mathbf{x}, t) - \psi(\mathbf{x}, t) \nabla \psi^*(\mathbf{x}, t)).$$



This problem can be solved in many ways. You can choose however you want to approach it. Below, I provide four different solutions using four different formalisms: wavefunctions, Dirac notation, propagators, and density matrices.



Note that this problem is unrealistic: real energy eigenfunctions cannot be strictly localized in space. However, this is a useful idealization for us to explore time evolution and other basic concepts. The goal is to flex our technology a little, even if not strictly necessary to solve this particular problem.

Solution in terms of Wavefunctions

Let us start with wavefunctions and solve this as you would in your undergraduate QM course.

a) By the standard unitary time evolution, $\psi(\mathbf{x}, t) = e^{-iEt/\hbar}\psi_1(\mathbf{x}, 0)$ in the energy eigenbasis. If initially $\psi(\mathbf{x}, 0) = 0$ for $\mathbf{x} \notin \Omega_1$, then at any later time t we have

$$|\psi(\mathbf{x}, t)|^2 = |e^{-iEt/\hbar}\psi_1(\mathbf{x}, 0)|^2 = |\psi_1(\mathbf{x}, 0)|^2 = 0 \quad \text{for } \mathbf{x} \notin \Omega_1.$$

These are called stationary states.

b) If the two regions do not overlap, then $\psi_1^*(\mathbf{x})\psi_2(\mathbf{x}) = \psi_1(\mathbf{x})\psi_2^*(\mathbf{x}) = 0$ for all $\mathbf{x}$. Already you should be convinced that the two wave functions do not interfere with each other. The time-evolved wave function is simply:

$$\psi(\mathbf{x}, t) = \frac{1}{\sqrt{2}} (e^{-iE_1 t/\hbar}\psi_1(\mathbf{x}) + e^{-iE_2 t/\hbar}\psi_2(\mathbf{x})),$$

and the probability density is trivial:

$$|\psi(\mathbf{x}, t)|^2 = \frac{1}{2} (|\psi_1(\mathbf{x})|^2 + |\psi_2(\mathbf{x})|^2 + 2\Re [e^{-i(E_1-E_2)t/\hbar}\psi_1(\mathbf{x})\psi_2^*(\mathbf{x})]) \quad (1)$$

$$= \frac{1}{2} (|\psi_1(\mathbf{x})|^2 + |\psi_2(\mathbf{x})|^2), \quad (2)$$

which is independent of time (as shown above).

c) If the regions overlap, then the interference term in the probability density does not vanish. Let us write it out after factoring out the phase from each wavefunction: $\psi_j(\mathbf{x}) = |\psi_j(\mathbf{x})|e^{i\phi_j(\mathbf{x})}$, $j = 1, 2$. Then,

$$\Re [e^{-i(E_1-E_2)t/\hbar}\psi_1(\mathbf{x})\psi_2^*(\mathbf{x})] = |\psi_1(\mathbf{x})||\psi_2(\mathbf{x})| \cos(\phi_1(\mathbf{x}) - \phi_2(\mathbf{x}) - \omega t), \quad (3)$$

where $\omega = (E_1 - E_2)/\hbar$. Thus, the probability density is

$$|\psi(\mathbf{x}, t)|^2 = \frac{1}{2} (|\psi_1(\mathbf{x})|^2 + |\psi_2(\mathbf{x})|^2 + 2|\psi_1(\mathbf{x})||\psi_2(\mathbf{x})| \cos(\phi_1(\mathbf{x}) - \phi_2(\mathbf{x}) - \omega t)), \quad (4)$$

which is clearly periodic in time with angular frequency ω and period $T = 2\pi/\omega = 2\pi\hbar/|E_1 - E_2|$.



Physically, the probability density starts to spread between the two regions as time goes by, due to the overlap between the two wavefunctions. Note that at very late times, $t \gg T$, the cosine term oscillates very rapidly, and the time-averaged probability density becomes

$$\langle |\psi(\mathbf{x}, t)|^2 \rangle_t = \frac{1}{2} (|\psi_1(\mathbf{x})|^2 + |\psi_2(\mathbf{x})|^2).$$

This is the same result as in part (b), where there was no overlap, and we effectively recover a stationary state at late times. This is caused by the loss of coherence between the two overlapping wavefunctions at very late times.

d) We calculated the probability density above. The question asks for the *current* density, which describes the flow of probability through some region. Let us now simplify the problem using the spherical symmetry implied in the question:

$$\psi_{1,2}(\mathbf{x}, t) \rightarrow \psi_{1,2}(r, t), \quad \nabla \rightarrow \partial_r \mathbf{r},$$

where $r = |\mathbf{x}|$ points radially outward. In this case, the probability current density reduces to a radial flow:

$$\mathbf{j}(\mathbf{x}, t) \rightarrow j(r, t)\hat{\mathbf{r}} = \frac{\hbar}{2mi} (\psi^*(r, t)\partial_r \psi(r, t) - \psi(r, t)\partial_r \psi^*(r, t))\hat{\mathbf{r}}.$$

Substituting the time-evolved wave function into this expression and simplifying a little:

$$j(r, t) = \frac{\hbar}{2mi} \left[\frac{1}{2} \left(\psi_1^*(r) \partial_r \psi_1(r) - \psi_1(r) \partial_r \psi_1^*(r) + \psi_2^*(r) \partial_r \psi_2(r) - \psi_2(r) \partial_r \psi_2^*(r) \right) \right. \quad (5)$$

$$\left. + \frac{1}{2} \left(e^{-i\omega t} (\psi_1^*(r) \partial_r \psi_2(r) - \psi_2(r) \partial_r \psi_1^*(r)) + e^{i\omega t} (\psi_2^*(r) \partial_r \psi_1(r) - \psi_1(r) \partial_r \psi_2^*(r)) \right) \right]. \quad (6)$$

Note that the first bracket vanishes since we are assuming the eigenfunctions are real.¹ The second bracket can be simplified using the product rule:

$$j(r, t) = \frac{\hbar}{2mi} \left[\frac{1}{2} \left(e^{-i\omega t} \partial_r (\psi_1^*(r) \psi_2(r)) - e^{i\omega t} \partial_r (\psi_2^*(r) \psi_1(r)) \right) \right] \quad (7)$$

$$= \frac{\hbar}{2m} \sin(\omega t) \partial_r (\psi_1(r) \psi_2(r)). \quad (8)$$

Thus, the probability current density flowing through region Ω_1 is

$$j(r, t) = \frac{\hbar}{2m} \sin \left(\frac{E_1 - E_2}{\hbar} t \right) \partial_r (\psi_1(r) \psi_2(r)).$$

We know this makes sense. Suppose there is an empty shell between the two spheres, i.e., $\psi_1(r) \psi_2(r) = 0$ for $R_2 < r < R_1$. Then, the current density vanishes everywhere in the spheres, i.e., there is no flow of probability between the two regions. Similarly, if the two spheres completely overlap, i.e., $R_1 = R_2$, then the current density is non-zero everywhere inside the sphere, until the probability density becomes homogeneous. Note that for homogeneous wavefunctions, i.e., $\psi_j(r) = A_j$ for $r \leq R_j$ and zero otherwise, $j = 1, 2$, the current density vanishes everywhere except at the boundary $r = R_2$ where $\psi_2(r)$ drops to zero discontinuously.



Again, the flow of probability at very late times averages to zero, since $\langle \sin(\omega t) \rangle_t = 0$. This is consistent with our earlier observation that the time-averaged probability density becomes stationary at late times.

Discussion in Dirac notation

Now we revisit the problem with a bit more abstract, but neater notation using Dirac kets and bras. We will be less thorough here since the steps are similar to those above. Let us define the states $|1\rangle$ and $|2\rangle$ corresponding to $\langle \mathbf{x}|1\rangle = \psi_1(\mathbf{x})$ and $\langle \mathbf{x}|2\rangle = \psi_2(\mathbf{x})$. Note that while it is true that $\langle 1|2\rangle = \int_{\mathbb{R}^3} d^3\mathbf{x} \langle 1|\mathbf{x}\rangle \langle \mathbf{x}|2\rangle = 0$, since we know these correspond to non-degenerate energy eigenstates, it may not be true that $\langle 1|\mathbf{x}\rangle \langle \mathbf{x}|2\rangle = 0$ for any given $\mathbf{x}$. The only information we are given is about the support of the wavefunction in position space (where they are localized with unrealistically sharp edges).

(a) We first introduce the projector onto region Ω_i :

$$\hat{\Omega}_i \equiv \int_{\Omega_i} d^3x |\mathbf{x}\rangle \langle \mathbf{x}|.$$

¹This is generally true for wavefunctions in 1D and real spherically symmetric potentials in 3D (spherical symmetry implies zero-angular momentum $\ell = 0$, reducing Schrödinger's equation for r to the same form as the 1D case). You can convince yourself of that by noting that the time-independent Schrödinger's equation is purely real in these cases.

If the two regions do not overlap, then

$$\hat{\Omega}_i |j\rangle = \delta_{ij} |j\rangle, \quad i, j = 1, 2.$$

(Note that at this point, the generalization to any number of regions is trivial.) We can also define the complementary projector $\hat{U}$ defined as (sorry for the wild notation here)

$$\hat{U}_i = \int_{\mathbb{R}^3 \setminus \Omega_i} d^3x |\mathbf{x}\rangle \langle \mathbf{x}|,$$

which projects onto the space outside region Ω_i . Note that by the completeness relation, we have $\hat{\Omega}_i + \hat{U}_i = \hat{I}$, where $\hat{I}$ is the identity operator.

The assumption that the particle is “initially in Ω_1 ” means that $\hat{\Omega}_1 |\psi(0)\rangle = |\psi(0)\rangle$ and $\hat{U}_1 |\psi(0)\rangle = 0$. We can ask: are these projectors constants of motion? In other words, do they commute with the Hamiltonian?² We can show that this is the case by calculating the commutator directly. Let us write out the commutator explicitly and make use of the complementarity between $\hat{\Omega}_i$ and $\hat{U}_i$:

$$[\hat{H}, \hat{\Omega}_i] = \hat{H}\hat{\Omega}_i - \hat{\Omega}_i\hat{H} = \hat{H}(\hat{I} - \hat{U}_i) - (\hat{I} - \hat{U}_i)\hat{H} = -[\hat{H}, \hat{U}_i].$$

Rearranging,

$$[\hat{H}, \hat{\Omega}_i] + [\hat{H}, \hat{U}_i] = 0 \implies [\hat{H}, \hat{\Omega}_i + \hat{U}_i] = [\hat{H}, \hat{I}] = 0,$$

which is trivially true. Thus, the projectors commute with the Hamiltonian, and are constants of motion. Therefore, if the particle is initially in region Ω_1 , it will remain there forever.

b and c) Time evolving the initial state $|\psi(0)\rangle$ gives

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} (e^{-iE_1 t/\hbar} |1\rangle + e^{-iE_2 t/\hbar} |2\rangle),$$

from which we can already calculate $\rho(\mathbf{x}, t) = \langle \psi(t) | (|\mathbf{x}\rangle \langle \mathbf{x}|) | \psi(t) \rangle$. But let us not worry about individual points in what follows and play a little with the whole regions instead. This will show some neat operations we can do with the “region projectors”.

The cross term $\langle 1|2\rangle$ is crucial to determine whether the wavefunctions overlap in the intersection region. In fact, it is non-zero only if there is some overlap between the two regions. Let us consider some region $\Omega_{12} = \Omega_1 \cap \Omega_2$ where there could be some overlap. This intersection can be assigned its own projector:

$$\hat{\Omega}_{12} = \hat{\Omega}_1 \hat{\Omega}_2 = \hat{\Omega}_2 \hat{\Omega}_1.$$

Also, by construction, our wavefunctions are restricted to a given Ω_i region, so it is useful to identify our identity operator as the sum of all projectors onto each region and its complement:

$$\hat{I} = \hat{\Omega}_1 + \hat{\Omega}_2 - \hat{\Omega}_{12}.$$

That is, the system must be in either region 1 or region 2, but we remove any double-counting from the overlap region. With this definition, the cross term is simply

$$\langle 1 | \hat{I} | 2 \rangle = \langle 1 | (\hat{\Omega}_1 + \hat{\Omega}_2 - \hat{\Omega}_{12}) | 2 \rangle.$$

By the way, Ω_{12} is the only operator that makes sense to talk about in the inner product between $|1\rangle$ and $|2\rangle$. To see this, we can write:

$$\hat{\Omega}_{12} = \hat{\Omega}_1 \hat{\Omega}_2 = (\hat{I} - \hat{U}_1)(\hat{I} - \hat{U}_2) = \hat{I} - \hat{U}_1 - \hat{U}_2 + \hat{U}_1 \hat{U}_2,$$

²You should know how to prove this equivalence.

or equivalently, $\hat{I} = \hat{\Omega}_{12} + \hat{\Omega}_1 + \hat{\Omega}_2 - \hat{\Omega}_1 \hat{\Omega}_2$, which means

$$\langle 1 | \hat{I} | 2 \rangle = \langle 1 | (\hat{\Omega}_{12} + \hat{\Omega}_1 + \hat{\Omega}_2 - \hat{\Omega}_1 \hat{\Omega}_2) | 2 \rangle = \langle 1 | \hat{\Omega}_{12} | 2 \rangle.$$

since $\hat{\Omega}_i |i\rangle = 0$ by definition. The exact time dependence of the probability density can be calculated explicitly as before, so we stop here.

d) With a spherically symmetric setup, our region projectors become rather simple:

$$\hat{\Omega}_i = \int_0^{R_i} dr \int d\cos\theta d\phi r^2 |r, \cos\theta, \phi\rangle \langle r, \cos\theta, \phi|, \quad i = 1, 2, \quad (9)$$

which projects onto all points inside a sphere of radius R_i . We can now see explicitly how the overlap region appears in the inner product:

$$\langle 1 | \hat{I} | 2 \rangle = 4\pi \int_{R_1}^{R_2} dr r^2 \langle 1 | r \rangle \langle r | 2 \rangle = 4\pi \int_{R_1}^{R_2} dr r^2 \psi_1^*(r) \psi_2(r),$$

which is a 1D integral over the radial coordinate only, since the angular parts integrate to 4π due to spherical symmetry.

As you showed in the first problem set, the probability current at a infinitesimal region around x location can be written in Dirac notation as

$$\hat{\mathbf{j}}(\mathbf{x}) = \frac{\hbar}{2mi} (|\mathbf{x}\rangle \langle \mathbf{x}| \hat{p} + \hat{p} |\mathbf{x}\rangle \langle \mathbf{x}|).$$

and you can ask if the probability current flowing through the surface of the regions Ω_i , which we can call $\partial\Omega_i$ and calculate as

$$\widehat{\partial\Omega_i} = \int d\cos\theta d\phi |R_i, \cos\theta, \phi\rangle \langle R_i, \cos\theta, \phi|, \quad (10)$$

where the fixed radius defines the surface of the sphere and the angular labels define a coordinate on it. Note that $\langle \mathbf{x}' | R_i \rangle = \delta(R_i - r') \delta(\cos\theta - \cos\theta') \delta(\phi - \phi')$. Now to calculate the flux through the surface region $\partial\Omega_i$, we can define the current operator defined over the spherical surface. To do that, define a new operator symmetrized between the current operator and the surface projector. The way to get something that projects onto the surface is to convolve the current operator with the surface projector. Assigning a superscript $'$ to the dummy integration variables to distinguish it from the coordinates of the $\partial\Omega_i$ surface, we have:

$$\begin{aligned} \hat{\mathbf{j}}_{\Omega_i} &= \int dr' d\cos\theta' d\phi' r'^2 (\widehat{\partial\Omega_i} \hat{\mathbf{j}}(\mathbf{x}') + \hat{\mathbf{j}}(\mathbf{x}') \widehat{\partial\Omega_i}) \\ &= \int dr' d\cos\theta' d\phi' \int d\cos\theta d\phi r'^2 (|R_i\rangle \langle R_i| \mathbf{x}' \rangle \langle \mathbf{x}' | \hat{p} + \hat{p} | \mathbf{x}' \rangle \langle \mathbf{x}' | R_i \rangle \langle R_i|) \\ &= 4\pi R_i^2 (|r\rangle \langle r| \hat{p} + \hat{p} |r\rangle \langle r|). \end{aligned}$$

where in the last step we use the delta functions and trivially integrated over the angular parts. which is the symmetrized product of the current operator and the region projector. The probability current flowing through the surface of Ω_1 is then

$$\mathbf{j}_{\Omega_1}(t) = \langle \psi(t) | \hat{\mathbf{j}}_{\Omega_1} | \psi(t) \rangle \quad (11)$$

$$= 4\pi R_i^2 \langle \psi(t) | (|r\rangle \langle r| \hat{p} + \hat{p} |r\rangle \langle r|) | \psi(t) \rangle \quad (12)$$

$$= \frac{4\pi\hbar}{2mi} 4\pi R_i^2 (\psi^*(r) \partial_r \psi(r) - \psi(r) \partial_r \psi^*(r)), \quad (13)$$

where we used the fact that $\langle \psi(t) | r \rangle = \psi^*(r, t)$ and $\langle r | \psi(t) \rangle = \psi(r, t)$, as well as the representation of the momentum operator in position space, $\hat{p} \rightarrow -i\hbar \partial_r$. Recall that

$$\langle r | \hat{p} | \psi \rangle = -i\hbar \partial_r \langle r | \psi \rangle = -i\hbar \partial_r \psi(r).$$

This result in Eq. (13) is exactly the same as what we obtained in the wavefunction approach, up to the $4\pi R_i^2$ factor that accounts for the spherical surface area. If the problem were not spherically symmetric, we would have to keep the angular integrals in Eq. (13) and integrate over the whole surface to get the total flux through it.

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By the way, using Stoke's theorem, you can relate this surface integral to a volume integral over the region Ω_1 itself, if you want to calculate the total flow of probability into or out of that region. Note that using the continuity equation, you can relate this flow to the time derivative of the probability of finding the particle inside region Ω_1 . More precisely,

$$\begin{aligned} \frac{d}{dt} P_{\Omega_1}(t) &= \frac{d}{dt} \langle \psi(t) | \hat{\Omega}_1 | \psi(t) \rangle = \int_{\Omega_1} d^3x \frac{\partial}{\partial t} \rho(\mathbf{x}, t) \\ &= - \int_{\Omega_1} d^3x \nabla \cdot \mathbf{j}(\mathbf{x}, t) = - \oint_{\partial\Omega_1} d\mathbf{S} \cdot \mathbf{j}(\mathbf{x}, t), \end{aligned} \quad (14)$$

but the last integral is exactly what we calculated above. So, the time derivative of the probability of finding the particle in region Ω_1 is equal to minus the probability current flowing through its surface.

Discussion with Propagators

Let us show that the propagator of the system does not allow the particle to leave region Ω_1 if it starts there. The propagator in the energy eigenbasis is given by

$$K(\mathbf{x}_2, t_1; \mathbf{x}_1, t_1) = \sum_n e^{-iE_n(t_2-t_1)/\hbar} \psi_n(\mathbf{x}) \psi_n^*(\mathbf{x}'),$$

where $\psi_n(\mathbf{x}) = \langle \mathbf{x} | n \rangle$ are the energy eigenfunctions.

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To prove this is easy, you just have to insert the completeness relation of the energy basis (assumed to be discrete here) twice into the definition of the propagator:

$$K(\mathbf{x}_2, t_2; \mathbf{x}_1, t_1) = \langle \mathbf{x}_2 | e^{-i\hat{H}(t_2-t_1)/\hbar} | \mathbf{x}_1 \rangle = \sum_{m,n} \langle \mathbf{x}_2 | m \rangle \langle m | e^{-i\hat{H}(t_2-t_1)/\hbar} | n \rangle \langle n | \mathbf{x}_1 \rangle,$$

and then use the orthonormality of the energy eigenstates.

It is then trivial to see how if the particle starts in region Ω_1 , it will remain there forever due to $\psi_i(\mathbf{x}) = 0$ for $\mathbf{x} \notin \Omega_i$. All other results follow as in the original solution.

Discussion with Density Matrices

Let us define the density matrix of the system at time t as

$$\hat{\rho}(t) = |\psi(t)\rangle \langle \psi(t)|.$$

If we split our states into the two regions, we can write

$$\hat{\rho}(t) = \sum_{i,j=1}^2 \rho_{ij}(t) |i\rangle \langle j|,$$

where $\rho_{ij}(t) = \langle i | \hat{\rho}(t) | j \rangle$ are the matrix elements of the density operator in the $\{|1\rangle, |2\rangle\}$ basis. The time evolution of the density operator is given by the von Neumann equation:

$$i\hbar \frac{d\hat{\rho}}{dt} = [\hat{H}, \hat{\rho}].$$

Calculating the matrix elements of both sides, we have

$$i\hbar \frac{d\rho_{ij}}{dt} = \langle i | [\hat{H}, \hat{\rho}] | j \rangle = \sum_{k=1}^2 \left(\langle i | \hat{H} | k \rangle \rho_{kj} - \rho_{ik} \langle k | \hat{H} | j \rangle \right).$$

But since $\hat{H} |i\rangle = E_i |i\rangle$, we have

$$\langle i | \hat{H} | j \rangle = E_i \delta_{ij},$$

so the equation simplifies to

$$i\hbar \frac{d\rho_{ij}}{dt} = (E_i - E_j) \rho_{ij}.$$

Because the two energy eigenvalues are different, $E_1 \neq E_2$, the off-diagonal elements evolve non-trivially in time:

$$\rho_{12}(t) = \rho_{12}(0) e^{-i(E_1 - E_2)t/\hbar}, \quad \rho_{21}(t) = \rho_{21}(0) e^{i(E_1 - E_2)t/\hbar},$$

while the diagonal elements are constant:

$$\rho_{11}(t) = \rho_{11}(0), \quad \rho_{22}(t) = \rho_{22}(0).$$

If the particle starts in region Ω_1 , then $\rho_{11}(0) = 1$ and $\rho_{22}(0) = \rho_{12}(0) = \rho_{21}(0) = 0$, so it remains in that region forever.

The rest of the problem follows as before, with the time dependence of the density matrix elements giving rise to the periodic behavior of the probability density and current when there is overlap between the two regions. One could also define region projectors and calculate expectation values using the density matrix formalism.